VOLUME v2.0

The Operator That Almost Exists

Eight new campaigns C9–C16, theorems T11–T18, and the boundary of the Hilbert–Pólya program after v1.0

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Introduction: what changed since v1.0

Version 1.0 built a complete numerical laboratory for the Hilbert–Pólya program: eight campaigns C1–C8, ten theorems T1–T10, 87 gates and three independent projections of a single gap H2. The v1 answer was honest and measured: the resonator is built and tuned (self-adjointness, GUE class, Dirac dynamics — YES); the zeros are encoded, not generated (spectral identity and primes-in-operator — NO, each with positive controls); and the prime weight $\Lambda(n)/\sqrt{n}$ provably lives in the heat trace of the zero operator (T6, $|\Delta K| \leq$ 1.2e-5).

The v1 roadmap named the directions able to move the main measured gap: nonlocal trace constructions (the T6 lesson), stitching the statistical and coding branches, a number-theoretic explanation of the $\alpha = 1/3$ zero modes, and extending the verified band. Version 2.0 executes all five items and adds five new measurements: blind campaigns, an anomaly hunt outside the GUE attractor, transferring the pipeline to Dirichlet L-functions, a quantum-information layer, and an open challenge with a public metric.

The honesty discipline is strengthened: in v2 the package uses NO external data file at all — the zeta zeros are computed by mpmath inside the package (the data/zeros/ cache of 800 zeros is bit-reproducible), and every metric convention is documented. Four errors and oversights were found-and-fixed in v2, including the Hadamard constant in the v1 shipped code; all are published in the honesty ledger (section 9).

The v2 summary in one sentence: **the operator almost exists** — the primitive with the prime weight in its trace is built exactly and made arbitrarily nonlocal (C9/T11); arithmetic entered the lattice spectrum for the first time through the counting law of zero modes (C11/T13); the zeros'-level result turned out to be a universal property of the L-function class (C15/T17); and every time the open branch passes through locality, not through the trace.

C9/T11. The nonlocal primitive: the trace knows the zeros exactly

The T6 lesson was that the heat trace of the operator stitched from the zeros carries the prime weight exactly: $K(x) = \Sigma\, e^{-x\gamma} = (1/\pi)\int_0^\infty Q(a)\sin(ax)\, da$, where $Q(a)$ unfolds into the prime comb $\Lambda(n)/\sqrt{n}$ through ζ'/ζ for $a > \frac{1}{2}$. Roadmap item R1 asked: can one build a family of operators whose trace carries this weight BY CONSTRUCTION, not through the lattice — and does that move r_EF out of the null cloud of gates?

The first result of C9 is theorem T11, and it flips the question. **Theorem T11 (heat-trace uniqueness).** If $\mathrm{Tr}\, e^{-xH} = (1/\pi)\int Q(a)\sin(ax)\, da$ for all $x > 0$, then $\mathrm{spec}\, H = \{\pm\gamma_n\}$. The proof is three lines: both sides are Laplace–Stieltjes transforms of measures on $(0,\infty)$; by uniqueness the measure is determined by the trace; since $(1/\pi)\int 2a\cdot\sin(ax)/(a^2+\gamma^2)da = e^{-x\gamma}$, the measure is the counting measure of the zero set. The architectural corollary: the prime weight in the trace is not decoration — it is the spectral measure itself; while nonlocality is achievable arbitrarily.

The C9 family realizes this measurably: N1 — $H_\xi = \mathrm{diag}(\pm\gamma)$ (control); N2 — the DCT twin $O\cdot\mathrm{diag}(\gamma)\cdot O^T$ with a deterministic dense orthogonal conjugation; N3 — the QR twin with a chaos seed independent of the zeros; N4 — truncated Berry–Keating xp (the outside-the-lattice candidate); N5 — the multiplicative kernel

Λ(lcm(m,n))/√mn with the prime weight in the kernel itself. The v2 convention (unfolded-vs-unfolded, documented in docs/v2) makes the control honest: an isospectral twin must give r_H7 = 1.000 — and it does.

Result: N2/N3 give r_H7 = 1.000000 and r_EF(fluct) = 0.999 — at PC0 level; the twin's spectral deviation from the zeros is 7e-13; the T6 trace verification on the internal zero cache reproduces to $|\Delta K| \le 1.8e\text{-}6$. In contrast, Berry–Keating (r_H7 = 0.047, z ≈ 0.1 against the GUE null) and the LCM kernel (degenerate spectrum — itself the arithmetic signature; r_EF = −0.96) stay in the null zones. Verdict: **the primitive exists, and the open bit-level branch is a wall of locality, not of the trace**.

C10/T12. Phase coding: hard co-location and its limit

Campaign C6 showed: the prime flow $\Phi_p = \frac{1}{2}\ln p$ composites exactly ($(\Phi_p,\Phi_q) \equiv \Phi_{pq}$, 1.35e-14), but the spectrum does not see the coding. Roadmap R2 demanded phase configurations OBLIGED to satisfy co-location. C10 solves this constructively and obtains a classification along the way.

Theorem T12 (classification). Any phase system Φ: primes → $\mathbb{R}/2\pi\mathbb{Z}$ with the identity $\Phi_{pq} \equiv \Phi_p + \Phi_q \pmod{2\pi}$ is a homomorphism from the free abelian group ⟨p⟩; it is arbitrary on generators, and its ln-like normal form is $\Phi_p = 2\pi\{\beta \ln p\}$. For the characteristic case $\Phi_p = \pi\cdot\chi_4(p)$ the system takes values {0, π} and turns out **gauge-trivial** in the lattice overlay channel: $e^{\pm i\pi} = -1$ flips every y-bond, and the transformation $\psi \to (-1)^y \psi$ removes the coding entirely (spectra coincide to 8.4e-15 — measured). The only non-gauge class is $\beta\cdot\ln p$.

Measurement: gates with a site-linked overlay Φ_p (prime ↔ site index; a row-uniform overlay is pure gauge and does not change the spectrum — also measured during debugging). The hard co-location condition holds at machine zero (4.4e-16) for ANY β — by construction, before the spectrum runs. The spectral answer: r_H7 ≤ 0.092 across the whole β-grid of 11 values (best p = 0.015, Bonferroni 0.20), the GUE class unchanged (⟨r⟩ = 0.58–0.63). Verdict: **universality erases even hard codings** — consistent with M3 of v1, but now on the class of systems for which co-location is an identity, not an approximation.

C11/T13. The arithmetic bridge: zero modes at α = 1/3

The T9 observation (v1): at α = 1/3 there are zero modes on tori L ≡ 0 (mod 4) — 18 at L = 12, 42 at L = 24. The roadmap demanded the bridge «arithmetic L ≡ 0 (mod 3) → Dirac index». C11 builds a clean laboratory: the Kähler–Dirac (staggered) operator in the γ_5-Hermitian form $H = \varepsilon\cdot D + m$ with Landau flux α; the torus requires $\alpha\cdot L \in \mathbb{Z}$ — at α = 1/3 this is exactly L ≡ 0 (mod 3): flux quantization is the geometric cause of the first arch of the bridge.

The full scan L = 6..48 (15 sizes, tolerance sweep 1e-6..1e-10) gives the law: **$N_0 = 2L - 6\cdot[L \equiv 0 \pmod{12}]$ for L ≡ 0 (mod 6), else $N_0 = 0$**; the Dirac index Σ sign(ε) equals −2 exactly on the subfamily L ≡ 0 (mod 12) and 0 on L ≡ 6 (mod 12). The decisive checks on sizes absent in v1: L = 36 → 66, L = 48 → 90 — both exactly on the law. The T9 observation is refined: L = 18 (≡ 2 mod 4) also carries 36 modes — the modifier is mod 12, i.e. the conjunction of mod 3 (flux) and mod 4 (staggered structure).

Additional measurements: the open boundary gives exactly one zero mode with index −1 for odd L ≡ 0 (mod 3) — an edge index; the α = 1/2 control — no zero modes at any L. This is the first genuine arithmetic-to-spectrum bridge inside the program: the number of zero modes of the lattice is read off the residues of L, and the index distinguishes two arithmetic subclasses. Proving the law through the Kähler–Dirac index is the main goal of v3 (N1 in the roadmap).

C12/T14. The argument principle band to T = 5000

The C2 machine (adaptive argument principle with the Turing bridge of arithmetics) is extended to T = 5000. The core is the Riemann–Siegel function: Z(t) = mp.siegelz at dps = 20 with the bridge to e^{iθ}ζ at machine precision (max |Δ| = 0.0 at six control heights). The on-line count: a coarse grid of 0.15 with refinement of small-|Z| intervals (catching close pairs) and 36-iteration bisection; the run is done in segments (0–5000 in 13 runs, ~60 minutes).

Result: **N_online(5000) = 4520** against the smooth count Ñ(5000) = 4520.08; S_max = 0.91 ≤ 2 (the Turing validity threshold of the block scheme); strip checks by the argument principle over the full strip σ ∈ [0,1] in three windows — 11 = 11, 10 = 10, 5 = 5; the v1 anchors (0, 3, 10, 29, 79, 649) are reproduced bit-wise. Gaps: [0.0908, 6.887], mean 1.103. The verified-RH band grew from 649 zeros to 4520 — an order of magnitude in height and sevenfold in count.

C13/T15 + C14/T16. Blind families and the attractor taxonomy

Five gate families absent from the v1 handle space (Nv, q, W, α, aniso, dalpha, mass, profile, signs, bc, twist): BF1 — purely imaginary x-hoppings (phase +π/2, TRS broken differently); BF2 — quasiperiodic vortex positions by the golden ratio, no RNG; BF3 — staggered columns of vortex dimers; BF4 — long-range hoppings ~$1/r^2$; BF5 — row-by-row flux dithering. Twenty gates, one pipeline, pipeline positive control PC0' = 1.000.

Result of the blind protocol: all 20 gates are in the null zone — |r_H7| ≤ 0.040, minimal p = 0.29, GUE class intact. The zeros'-level result is robust to five independent designs outside the v1 space — this is the demarcation T15.

The anomaly hunt (C14/T16) — a systematic scan of 121 configurations with an honest taxonomy: 97 configurations are the full GUE class; 18 are the integrable clean lattice (Nv = 0, Poisson/critical class — expected from C8); 6 are intermediate statistics of a weak vortex cloud (Nv ≤ 8 at q = 0.5, ⟨r⟩ = 0.47–0.53). No rank outliers (|r_H7| ≤ 0.153). Conclusion: **the GUE attractor is global for all vortex-carrying configurations**, but its basin has a measurable structure in (Nv, q, W) — a map of the approach to universality, not a sensation outside it.

C15/T17. Dirichlet L-functions: the rank-metric resolution limit is universal

The main question of plan item 12: is the measured rank-metric resolution limit universal, or a defect of zeta specifically? C15 transfers the C1–C4 + C6 pipeline to two quadratic characters: χ_5 (even, mod 5) and χ_7 (odd, mod 7).

The graph branch: $Z_\chi(t) = e^{i\theta_\chi(t)}L(½+it,\chi)$ is real on the line to 5e-35 (K2 analog); bisection gives 68 zeros ≤ 120 for χ_5 and 74 for χ_7 — matching the smooth counts 67.8 and 74.5; a sample strip check by the argument principle confirms the count (3 = 3 in two windows). The ψ_χ ladder: the twisted sieve $\Sigma\chi(n)\Lambda(n)$ against the zero reference — correlations 0.902 and 0.949 (M2 analog with a positive control).

The trace branch — theorem T17: **$K_\chi(x) = \Sigma\, e^{-x\gamma_\chi} = (1/\pi)\int Q_\chi(a)\sin(ax)da$, $Q_\chi(a) = \Lambda'/\Lambda(½+a,\chi) - B_\chi - C_0^\chi$ carries the twisted weight $\chi(n)\Lambda(n)/\sqrt{n}$**. The derivation repeats the Hadamard scheme for the entire function Λ_χ; along the way the exact identity $C_0^\chi = \Lambda'/\Lambda(½) - B_\chi$ was found (an analog of $B = -C_0$ from v1). Verification: $|\Delta K| \le$ 9.6e-6 (χ_5) and 6.1e-6 (χ_7) — v1-level precision, with the prime part χ-twisted.

The rank projection: the same gates against L-function zeros — r_{H7} = 0.035–0.164, the null zone. Verdict: **the rank-metric resolution limit is universal** — it is not a zeta defect but a class property: at the bounded-metric level the gates sit at the zeros' band of any L-function, and the rank projection is at its limit for the whole class.

C16/T18. The quantum-information layer

The third measurement of the resonator — through entanglement and complexity. The correct object for one-particle Hamiltonians is the fermionic entanglement of the Slater Fermi sea: fill the n lowest orbitals, the correlation matrix $C_A = P_A(\Sigma\psi\psi^\dagger)P_A$, and the spectrum $\lambda \in [0,1]$ gives $S_{vN} = -\Sigma\lambda \ln\lambda - \Sigma(1-\lambda)\ln(1-\lambda)$ (the Peschel protocol).

Q1: the gate's Fermi sea is weakly entangled — $S_A \approx$ 12–16 nats against the Haar baseline 160–220 ($z >$ 100): **locality suppresses the volume-law entanglement** — the quantum-information echo of the wall of locality C9/T11. Q2: the mutual information of the halves carries no zero signature — the correlation of the MI profile with the zero window is −0.018 against −0.121 random (both in the null cone). Q3: the spectral form factor on unfolded windows — the gate and GUE slopes match within the spread, the zero window is more rigid. Verdict: **entanglement and complexity erase arithmetic along with spectral statistics** (T18).

The v2 honesty ledger: found-and-fixed

The v1 discipline («statuses are not inflated, all constants are derived») is continued; v2 publishes four findings:

1. **B_HADAMARD** in the v1 shipped code contained ln(2π) = 1.8379 instead of ln(4π) = 2.5310. The correct value B = −0.0230957 is confirmed by the committed v1 verdicts — the error was in the file constant only, not in the results. Fixed with a code comment.

2. **The T9 observation of v1 is refined**: the full C11 law — $N_0 = 2L - 6\cdot[L \equiv 0 \pmod{12}]$ — shows that modes exist on all L ≡ 0 (mod 6), including L = 18 (≡ 2 mod 4); the v1 formulation «tori L ≡ 0 (mod 4)» described only the index −2 subfamily.

3. **The χ4 coding is gauge-trivial** — phase systems with values {0, π} are removed by the transformation ψ → (−1)^y ψ; the spectral coincidence 8.4e-15 is measured. This is a corollary of T12 and an honest constraint of the coding class.

4. **The r_EF metric of v2** — on detrended fluctuations: the raw v1 correlation catches the common trend of two smooth curves (any smooth D(u) gives r ≈ 0.99). The GUE null is wide (±0.8) and stated explicitly; the isospectral control must give r_fluct = 0.999 — and does.

Additionally: the T6 check passed n instead of ln n to K_formula; the pole a = 0.5 fell on the spline grid; the environment's background processes do not survive between calls — the C12 run was done in segments with a later assembly. All documented in code and verdicts.

Explorer, Challenge and reproducibility

Bridge Explorer (explorer/streamlit_app.py, `streamlit run`) — a live tool: turn the gate handles → the 576×576 spectrum is rebuilt in place → three projections of the gap + the C9–C16 verdicts + the bridge map. Every live number uses the same protocol as the campaigns.

The open challenge (challenge/CHALLENGE.md + score_submission.py): the public metric — demonstrate r_H7 ≥ 0.7 under the v2 protocol with passing positive controls and an intact GUE class. The scorer is blind, rebuilds the gate from the spec itself, and saves the full report to challenge/submissions/ regardless of the outcome. The leaderboard is open.

Reproducibility: IVP01–IVP18 — 18/18 PASS in one command; pytest 21/21; the verdict-replay campaigns re-read the committed data in seconds; the fresh runs (C9–C16) are self-contained and need nothing beyond numpy/scipy/mpmath. The zeta zeros are an internal cache with the mpmath version fixed in the verdicts.

The v3 roadmap

N1. **Proving the T13 law**: explain $N_0 = 2L - 6\cdot[L \equiv 0 \pmod{12}]$ through the Kähler–Dirac index and the spectral sequence of the torus with flux 1/3 — the first full «theorem from arithmetic» inside the lattice.

N2. **Other characters and forms**: χ mod 3, 11, 13 and modular forms (level 11) — how far does the T17 universality reach.

N3. **Stitching through the index**: the Kähler–Dirac operator itself as a candidate gate — does its heat trace carry the prime weight at α = 1/3?

N4. **The band T = 50000** with the same segment machine (~40× the C12 time).

N5. **Quantum response to the index**: does MI see the phase transition $N_0 > 0 \Leftrightarrow L \equiv 0 \pmod 6$ in the C16 protocol.

N6. **The challenge**: live submissions; the first accepted result — a joint paper.

The v2.0 verdict

The operator that almost exists exists exactly as far as the trace allows: with the prime weight in it — exactly (T11); with arbitrary nonlocality — exactly; with prime co-location in the phases — exactly (T12). It refuses to exist locally: none of the 227 measured configurations of five families and three classes left the null zones, and the approach to GUE itself turned out to be structured (T15, T16). Arithmetic manifested in the lattice spectrum for the first time — but where nobody looked: in the count of the Dirac-index zero modes (T13). And the gap measured on zeta turned out to be a property of the whole L-function class (T17). The continuation is in the v3 roadmap.